MIL-D-14543B
7 August 1973
SUPERSEDING
MIL-D-14543A(MC)
13 January 1967

MILITARY SPECIFICATION

DIE, RETHREADING: AXLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. This specification covers one type hinged, clamp type axle rethreading die for cutting and renewing bruised, battered, crossed or rusty threads.

2. APPLICABLE DOCUMENTS

*2.1. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables For Inspection By Attributes.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

FSC 5136

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NATIONAL BUREAU OF STANDARDS

Handbook H28-Screw-Thread Standards for Federal Services

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

*3.1 Material. Material shall be as specified herein. Material not definitely specified shall be in accordance with good commercial quality for the intended purpose.

*3.2 Design. The rethreading die shall be new, convenient to operate, and entirely adequate for the intended purpose. The tool shall be of split type design and hinged so that the dies may be opened and clamped around the undamaged portion of an axle thread and revolved to renew the damaged thread section. The rethreading die shall incorporate not less than six different thread cutting dies. The dies shall consist of the following six sizes of right hand threads conforming to the American National form as specified in H28:

Diameter (inches)	Threads per inch
5/8	18
3/4	16
7/8	14
1	14
1-1/8	12
1-1/4	12

The rethreading die shall consist of the following principal component parts.

* 3.2.1 Die jaws. The rethreading die shall consist of two forged steel die jaws. Each jaw shall contain a 180° sector of each of the different size thread cutting dies specified in 3.2. The die sectors shall be an integral part of the jaws or shall be fastened securely to the jaws and shall be set in a line so that the jaw will serve as a lever for rotating each die about its axis. The jaws shall be attached at one end by a hinge arrangement and clamped at the other end. The thread cutting portion shall have a Rockwell hardness of from 55 to 65 on the C scale to a minimum depth of 0.010 inch. A guide pin or comparable device shall be secured to one of the jaws at the clamp end to provide an accurate alignment of the die halves when they are clamped together in the cutting position. Each cutting die shall have not less than four equally spaced notches at least 5/64 inch wide and extending beyond the outside diameter of the thread, to provide chip clearance when recutting threads.

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*3.2.2 Hinge pin. A hinge pin shall be provided so that the die jaws can be opened and the tool installed around the axle shaft threads or removed from the axle shaft threads. The hinge pin shall be of adequate size to maintain alignment of the die jaws when in the rethreading position on the axle shaft permitting no side to side movement of the die jaws. The hinge pin shall be designed to prevent its dropping out.

*3.2.3 Clamp. A clamping device shall be provided at the end of the die jaws opposite the hinged end. The clamping device shall be capable of securely positioning and holding each of the dies in the rethreading position on its respective axle shaft thread. The clamping device shall maintain accurate positioning of the die jaws throughout the thread renewal operation.

*3.3 Construction. Construction shall be sound and free from any characteristics or defects that may render the rethreading die unsuitable or inefficient for the intended purpose. The rethreading die shall be complete so that when installed, it can be used for any operation for which designed.

*3.3.1 Threads. All threaded parts shall comply with H28.

*3.3.2 Surfaces. All forgings, molded parts, stampings, and welded parts shall be thoroughly cleaned and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smoothed and all edges shall be rounded or beveled. All bearing surfaces between matching parts shall be finished to the necessary tolerances. Other outside surfaces of unpainted parts shall be suitably finished.

*3.4 Performance. The axle rethreading die shall be a hand operated tool capable of restoring damaged threads on axles used on Military vehicles without removal of the axle shaft from the axle housing.

*3.5 Finish. All surfaces of the rethreading die, except the working surface of the dies, shall be finished with a suitable plating or coating to resist corrosion, except that cadmium plating shall not be used.

*3.6 Identification. The rethreading die shall be marked with the manufacturer's name or readily identifiable trademark, along with the size of the die.

*3.7 Handbook of instructions. Unless otherwise specified (see 6.2), one handbook of instructions shall be included with each rethreading die supplied under this specification. When specified (see 6.2), the supplier shall furnish a preliminary copy of the handbook of instructions for approval by the contracting officer prior to publication and delivery.

*3.8 Workmanship. The rethreading die shall be well finished, clean, free from burrs and all rough or sharp edges shall be smooth. The

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finished rethreading die shall be free from all defects and imperfections which may impair its serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification when such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

*4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Quality conformance inspection (see 4.3).

(b) Preparation for delivery inspection (see 4.6).

*4.3 Quality conformance inspection. The quality conformance inspection shall consist of the examination in 4.4 and the tests in 4.5.

*4.3.1 Sampling. Sampling for examination and test shall be performed in accordance with MIL-STD-105. For purposes of sampling, an inspection lot shall consist of all rethreading dies made with the same materials, components, essentially under the same conditions and offered for acceptance and delivery at one time. The inspection levels, for determining the sample size and the acceptable quality level (AQL) expressed in defects per hundred units (DHU), shall be as follows:

	Inspection Level	AQL's
Examination	II	2.5 Major 4.0 Minor
Test	S-4	4.0

*4.4 Examination. The rethreading dies shall be examined for the following defects:

Major

101. Size and thread form of die not as specified
102. Damaged thread cutting die, affecting operation
103. Failure of die jaws to open, close or mate properly
104. Clamping device not capable of securely clamping jaw

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Minor

201. External surfaces not finished
202. Corrosion
203. Handbook omitted
204. Marking illegible or not as specified.

*4.5 Tests. Tests shall be as follows:

*4.5.1 Rethreading. Six vehicle axles, or equivalent test specimens, having ends threaded to corresponding sizes specified in 3.2, shall have 15 to 30 percent of the threads purposely damaged. The threads shall be damaged to such an extent that the corresponding nut cannot be threaded on the axle without causing additional damage to the threads of the axle or nut. The proper rethreading die shall be clamped in position around the undamaged portion of the axle thread and revolved so as to recut the damaged portion. The corresponding nut shall then be mated satisfactorily on the axle. After each axle rethreading test, the split die used shall be visually examined for damage to the thread cutting impression. Failure of the rethreading die to operate properly shall be cause for rejection.

*4.5.2 Functional. Six vehicle axles, or equivalent test specimens, having ends threaded to corresponding sizes specified in 3.2, shall be inspected and checked against thread gages. Each split die of the rethreading die shall then be clamped into operating position on the proper vehicle axle. The axle shall have no damaged threads, but the tool shall be revolved to simulate rechasing damaged threads. The tool shall be removed and the axle threads visually examined for evidence of tool malfunction causing damage to the axle threads. Any damage caused by the rethreading die, to the axle, shall be cause for rejection.

*4.5.3 Hardness. Each thread cutting die, corresponding to those specified in 3.2, shall be Rockwell hardness tested as near the cutting threads as possible to determine compliance to 3.2.1.

*4.6 Preparation for delivery inspection. An inspection shall be made to determine compliance with the preservation, packaging, packing and marking requirements in Section 5.

5. PREPARATION FOR DELIVERY

*5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be level A, B or C of PPP-P-40 as specified (see 6.2).

*5.2 Marking. Marking shall be in accordance with PPP-P-40.

conformance to 1.2.P.1.

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6. NOTES

*6.1 Intended use. The rethreading die covered by this specification s intended to be used for restoring damaged axle threads on military ehicles.

*6.2 Ordering data. Purchasers should exercise any desired options ffered herein, and procurement documents should specify the following:

*6.2.1 Procurement requirements.

(a) Title, number, and date of this specification

(b) When number of handbooks required is other than one (see 3.7)

(c) When preliminary copy of handbook is required (see 3.7)

(d) Level of preservation, packaging and packing required (see 5.1).

*6.2.2 Contract data requirements. Data conforming to Marine Corps ata Item Description DI-M-4701 will usually be required for delivery n connection with this specification for Marine Corps procurements. or other than Marine Corps procurements, the cognizant procuring ctivity will provide comparable Data Item Descriptions. When so equired, such data will be specified for delivery on a DD 1423 ncluded in the contract.

*6.3 Marginal notations. The margins of this specification are arked with an asterisk to indicate where changes (additions, modifications, orrections, deletions) from the previous issue were made. This was done s a convenience only and the Government assumes no liability whatsoever or any inaccuracies in these notations. Bidders and contractors are autioned to evaluate the requirements of this document based on the ntire content irrespective of the marginal notations and relationship o the previous issue.

Custodians:

Army - GL
Navy - MC
Air Force - 84

User activities:

Army - CD
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Preparing activity:
Navy - MC
Project No. 5136-0033

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